

AI 4 STARTUPS

AI & DATA IN AFRICAN AGRICULTURE

From Use Cases to Real-World Impact – How startups are translating data and AI into operational, scalable solutions across the agricultural value chain.

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ABBREVIATIONS

PROGRAM & INSTITUTIONS

AI4SU	AI 4 Startups
EF	Expertise France
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
Data Gov	Data Governance in Africa Initiative
ANACIM	National Agency for Civil Aviation and Meteorology (Senegal)
BNEDT	Bureau National d'Études Techniques et de Développement
CMS	Cocoa Management System

REGULATORY & POLICY

AUDPF	African Union Data Policy Framework
DPO	Data Protection Officer
DSA	Data Sharing Agreement
EUDR	EU Deforestation Regulation
ODPC	Office of the Data Protection Commissioner

FINANCE & BUSINESS

B2B	Business-to-Business
DFI	Development Finance Institution
GDP	Gross Domestic Product
LTV	Lifetime Value
MFI	Microfinance Institution
MVP	Minimum Viable Product
POC	Proof of Concept
RFM	Recency, Frequency, Monetary value
SKU	Stock Keeping Unit
US\$	United States Dollar

TECHNOLOGY & AI

AI	Artificial Intelligence
API	Application Programming Interface
ASR	Automatic Speech Recognition
IoT	Internet of Things
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
TTS	Text-to-Speech

DIGITAL CHANNELS

IVR	Interactive Voice Response
SMS	Short Message Service
USSD	Unstructured Supplementary Service Data

REPORT STRUCTURE

UC	Use Case
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ALL PROGRAM RESOURCES

Open-source models, datasets, partner publications, and documentation are accessible on the AI4SU platform

ai4su.digital-africa.co



EXECUTIVE SUMMARY

AI4SU is a Digital Africa program implemented with Expertise France under the Data Governance in Africa Initiative. Through startup engagement, co-design workshops, and agritech-focused hackathons, the program tested how data sharing, open-source AI development, and applied governance frameworks can unlock value for African agritech startups. This report draws on the Briter Intelligence ecosystem mapping (2026), the Data Without Borders policy consultations (Data 354 / Digital Africa, 2026), and Digital Africa's operational program expertise.

8 AFRICAN MARKETS COVERED* **60+** STARTUPS ENGAGED **30+** WORKSHOPS ORGANIZED WITH STARTUPS **8** HACKATHONS CONDUCTED **24** OPEN SOURCE AI MODELS PRODUCED **9** USE CASES ACROSS AGRICULTURAL VALUE CHAIN

**Senegal, Côte d'Ivoire, Ghana, Kenya, Rwanda, Tanzania, South Africa, and Botswana*

CORE FINDING

The main bottleneck for AI in African agriculture is not algorithmic capability, but the availability, quality, governance, and operational usability of data. Across use cases, startups struggle less with model design than with collecting, structuring, accessing, validating, and deploying data in fragmented agricultural environments.

THE DATA CHALLENGE LOOP

Five interconnected challenges shape what startups can build and how fast they can scale



STRUCTURING DATA

Data exists but is non-digitised, unstandardised, or undocumented. Startups inherit ecosystem-level fragmentation and compound it through their own technical debt.



COLLECTING DATA

Granular agricultural data is scarce and expensive to produce. The cost of proprietary collection creates data moats that reinforce ecosystem silos.



SHARING DATA

Incentives, trust deficits, and legal complexity default the system to non-sharing — even when all parties would benefit from collaboration.



EXPLOITING DATA

Open data does not mean exploitable data. The gap between data access and value creation falls disproportionately on startups with the fewest resources.



GOVERNING DATA

Regulatory uncertainty, not regulation itself, is the binding constraint. Startups need comprehensible rules, not more rules.

These challenges form a reinforcing loop: weak structuring limits usability, which increases collection costs, which forces proprietary strategies, which reduces sharing, which fragments the ecosystem — and the cycle continues. Breaking it requires coordinated intervention across all five dimensions.

MAIN INSIGHTS

1. **Most use cases are data-driven before they are AI-driven**
Only a few use cases — crop monitoring, remote sensing, and voice-based interfaces — rely on AI as a technically central component. Many others depend more on rules, heuristics, proxy indicators, and domain expertise. This distinction matters: overstating AI's role creates misaligned expectations from investors, partners, and policymakers.
2. **Embedded models outperform standalone AI products**
Credit scoring, insurance, advisory, and market information become viable when embedded in broader services such as input finance, logistics, lending platforms, or agribusiness workflows. Standalone AI tools struggle to monetise; the surrounding business model matters more than the model itself.
3. **Value creation and value capture are structurally misaligned**
Many solutions create value for smallholder farmers, but farmers are rarely the paying customers. Value is captured by lenders, insurers, input providers, exporters, and platforms. This misalignment is not only a market failure to correct; it is also an operating reality that determines which use cases are commercially scalable and which require blended or public finance.
4. **Scaling is an operations problem, not a technology problem**
Agricultural data, farmer behavior, logistics, partnerships, and regulation are highly local. Cross-country scaling requires rebuilding data pipelines, distribution channels, and institutional relationships in each new market. Replicability is more constrained than technology narratives suggest — and investors who underestimate this consistently underperform.
5. **Data governance is a business condition, not only a compliance issue**
Consent, explainability, data-sharing agreements, platform dependency, and data ownership directly affect trust, access to data, and scalability. Governance failures do not just create legal risk — they block the data flows that startups need to build and improve their products.
6. **Traceability is becoming a market access gatekeeper**
Regulatory pressure — particularly the EU Deforestation Regulation — is transforming traceability from a competitive advantage into a condition of export market access. The strategic question for African producers: will the resulting data infrastructure generate sovereign data assets, or will it primarily serve importing market compliance systems?
7. **Farmer accessibility conditions every other use case's impact**
Voice interfaces, SMS, USSD, and conversational AI are not secondary features — they determine whether technically sound solutions actually reach the farmers who need them. The local language data gap is as binding as the agricultural data gap, and building it is a deliberate investment, not a byproduct of deploying generic models.

OVERALL CONCLUSION

AI can support African agriculture, but its impact depends on the systems around it: usable data, local calibration, trusted delivery channels, viable business models, and clear governance. The most promising startups are not simply building AI models — they are building operational systems that turn fragmented agricultural data into useful, trusted, and scalable services.

MAIN REPORT SOURCES

- Unlocking the Data Ecosystem for African startups report (Briter, 2026)
- Data Without Borders policy brief (Data 354 / Digital Africa, 2026)
- Use case documentation and hackathon challenge description

All program resources on

ai4su.digital-africa.co



I. CONTEXT & SCOPE

A. The AI4Startups program

The Data Governance in Africa Initiative supports the African Union and its member states to form development-oriented and human-centric data regulation, data use, and data infrastructure at continental, regional, and member state levels. The Initiative provides concrete support for a single digital market in Africa by harmonising regulations on data, creating cross-border data flows, and promoting investments in secure and sustainable data infrastructure facilities. It is funded by €60 million from the European Union and five of its Member States (Belgium, Estonia, Finland, France, and Germany) under the Global Gateway strategy, and implemented from 2023 to 2026 by GIZ, Enabel, and Expertise France as implementing partners, and Digital Africa, ESTDEV, and HAUS as grantees.

Within this initiative, the AI 4 Startups (AI4SU) program – implemented by Digital Africa in collaboration with Expertise France – takes a startup-centred approach to data governance and AI development with and for Agritech startups. Rather than working top-down on regulatory frameworks, the program starts from the concrete data challenges that African startups face when building AI-driven products. Its objective: unlock the value of data sharing with and for startups, through structured hackathons, open-source model development, and applied data governance frameworks.

Over the course of the program, AI4SU engaged more than sixty startups and organised around thirty workshops to identify, co-design, and refine data challenges suited to hackathon-based AI development. This process culminated in eight hackathons with agritech startups across multiple African markets, each structured around a specific data challenge – from soil nutrient prediction to crop classification, credit scoring to agricultural speech recognition. The hackathons produced 24 open-source AI models, now available to the broader ecosystem. Beyond these models, the hackathon process itself served as a practical test bed for data sharing: every challenge required negotiating data-sharing agreements, structuring IP protection, anonymising sensitive datasets, and navigating cross-border data transfer constraints.

All program resources – open-source models, datasets, documentation, and partner publications – are accessible on the AI4SU platform (ai4su.digital-africa.co), which serves as the central knowledge hub for the program. Beyond these publications, this report draws on Digital Africa's own operational expertise gained through the design and management of the AI4SU program: the co-design methodology developed with startups, the structuring of data-sharing agreements, and the practical lessons learned from running hackathons as applied data governance experiments.

The report draws in particular on two partner publications, both accessible on the platform:

- Unlocking the Data Ecosystem for African startups report (Briter): covering 250+ AgTechs, 250+ funders, and 70+ data policies across 8 African markets
- Data Without Borders policy brief (Data 354 / Digital Africa): synthesising multi-stakeholder consultations on data access barriers and solutions

B. Agriculture, AI, and the data bottleneck

The program focuses on agricultural technology – with openings to climate, energy, and food systems – for four reasons that make the sector uniquely suited to testing data-driven approaches in African contexts.

Data richness and diversity.

Agriculture generates a wide range of data types – soil health, weather patterns, satellite imagery, market prices, farmer profiles, transaction histories – each with different collection methods, formats, and access constraints. This diversity makes agriculture a demanding but representative test case for the data challenges that startups face across sectors.

Open data potential.

Compared to sectors like fintech or healthtech, where data is overwhelmingly proprietary and regulated, agriculture offers more pathways to open and shared-data: public weather stations, satellite imagery, government crop statistics, research institution datasets. The gap between this theoretical availability and actual usability is precisely what the program set out to investigate.

Public policy relevance.

Food security, rural employment, export competitiveness, and climate adaptation are policy priorities across all eight study countries. Data-driven agricultural solutions operate at the intersection of these agendas, giving the program's findings direct relevance to policymakers and development partners.

Variety of use cases with high impact.

The nine use cases analysed in this report span credit scoring for unbanked farmers, crop insurance calibration, yield prediction, supply chain traceability, market price forecasting, and voice-based advisory in local languages – demonstrating that data-intensive agriculture is not a single application but a spectrum of solutions with different data requirements and constraints.

AI as an opportunity – and a stress test for data ecosystems.

Across Africa, AI is capturing growing attention from startups, investors, and policymakers. In agriculture, the opportunity is tangible: AI-driven models can extract actionable insights from satellite imagery, predict crop yields from sparse historical data, or deliver personalised advisory in local languages at a scale that traditional extension services cannot match. But AI is also the most data-demanding form of digital innovation – and in contexts where data is scarce, fragmented, and poorly governed, it exposes every weakness in the data ecosystem.

This is precisely what makes AI a useful lens for this report: not as an end in itself, but as a stress test that reveals where data infrastructure, sharing mechanisms, and governance frameworks need to be strengthened.

C. Geographic scope

8 COUNTRIES COVERED

Senegal, Côte d'Ivoire, Ghana, Kenya, Rwanda, Tanzania, South Africa, and Botswana.

This selection responds to four complementary criteria:

Geographic and climatic diversity.

The eight countries span Sahelian, tropical humid, and temperate zones — a range that matters directly for agricultural data, where soil types, crop cycles, and weather patterns vary fundamentally across regions. Solutions built on Kenyan highland data do not transfer automatically to Senegalese groundnut basins.

Variable startup ecosystem maturity.

The Briter Intelligence diagnostic framework (2026) classifies the eight markets across four maturity levels — from nascent (Botswana) through activation (Senegal, Côte d'Ivoire, Tanzania) and market-responsive (Ghana, Rwanda) to scaling (Kenya, South Africa). This gradient allows the report to analyse how data challenges manifest differently depending on ecosystem development, and to identify which interventions are relevant at which stage.

Diverse data governance approaches.

The eight countries represent a range of regulatory strategies: data localisation requirements (Rwanda, Tanzania), right-to-explanation frameworks (Ghana, Kenya), DPO registration mandates (Côte d'Ivoire, South Africa), and AI-specific regulatory sandboxes (Rwanda, Kenya). This diversity is essential for analysing how governance frameworks affect startup operations in practice.

Regional hub logic and cross-border potential.

The selection creates three regional clusters — West Africa (Senegal, Côte d'Ivoire, Ghana), East Africa (Kenya, Rwanda, Tanzania), and Southern Africa (South Africa, Botswana) — enabling analysis of cross-border data flows, regional harmonisation dynamics, and hub effects.

This structure also creates synergies with other Data Governance in Africa components, particularly GIZ's work on regulatory harmonisation in several of these markets.

II. DATA CHALLENGES FOR STARTUPS

Across the African agritech landscape, a consistent finding emerges: the primary bottleneck for AI-driven agricultural innovation is not algorithmic capability but data. Startups building credit scoring models, insurance products, market intelligence tools, or traceability systems all confront a common set of structural data challenges that shape what can be built, how fast it can scale, and who captures the value.

This section maps five core data challenges faced by startups across the data value chain – from structuring and collecting to sharing, exploiting, and governing data. It draws on three complementary sources: the ecosystem mapping conducted with Briter Intelligence across eight African markets (Briter, 2026), direct engagement with startups through hackathon design and preparation, and the multi-stakeholder consultations synthesised in the Data Without Borders policy brief (Data 354 / Digital Africa, 2026). These challenges are deeply interconnected: poorly structured data limits sharing, constrained access limits exploitation, and regulatory uncertainty affects all stages. While they involve multiple stakeholders, startups are disproportionately impacted due to their limited resources and need for rapid iteration.



A. Structuring data: *Data availability does not equal data usability*

The absence of shared taxonomies, heterogeneous formats across institutions and countries, and limited interoperability create a fragmented data landscape that no single actor can fix. But startups do not only suffer from these ecosystem-level constraints – they also compound them through their own structuring failures, driven by lack of data engineering capacity, time pressure, and the trade-offs inherent in early-stage development.

- Fundamental agricultural data often remains non-digitised – paper records, oral knowledge, unstructured field observations that have never entered a digital pipeline.
- Where data exists digitally, conflicting standards between government departments create significant hurdles: without shared naming systems or common IDs for farmers and land, startups must invest substantial resources to make data from different sources work together (Briter, 2026).
- Adoption of global standards like AgroVoc (FAO) remains limited – creating inconsistencies where the same crop or input may be classified differently across systems. Kenya and South Africa have begun implementing interoperability standards; Rwanda leverages its digital ID infrastructure for farmer data matching; Senegal and Ghana are in earlier stages (Briter, 2026).
- At the startup level, funding and time pressure push teams to adopt minimal data structuring at MVP stage – no schema, no metadata, no documentation – creating a technical debt that becomes a wall at scaling.
- Data cleaning is a consistently underestimated cost. During hackathon preparation, significant effort was invested in cleaning, anonymising, and formatting startup data before it could be used for model development – revealing how far raw startup datasets are from being AI-ready.

- The shortage of data engineering profiles internally means structuring is often deferred – startups collect first, plan to structure "later," and rarely catch up.
- Jokalante's Kallaama project illustrates the challenge at its most fundamental: creating 125 hours of transcribed, annotated agricultural speech in local languages required systematic linguistic structuring – segmentation, transcription, quality validation, and metadata annotation – before any model development could begin.

IMPLICATION

Data structuring should be explicitly funded as a standalone investment – it is not a byproduct of data collection, and its absence blocks every downstream use. Shared agricultural data standards (taxonomies, formats, identifiers) would reduce cross-platform and cross-border integration costs, particularly for startups expanding to new markets.

CHECK USE CASE 9 : KALLAAMA / JOKALANTE

building annotated speech datasets in local languages required systematic linguistic structuring before any AI model development could begin.



B. Collecting data: *Data collection is not just a technical challenge but a strategic trade-off between cost, control, and competitiveness.*

Data scarcity is a defining feature of African agricultural markets. Public datasets, where available, tend to rely on national averages that fail to capture the plot-level variability required for precision agriculture or localised credit scoring. For startups, this scarcity forces a costly "build your own data" model with structural consequences on growth and competitive dynamics.

- Across the 8 study countries, over 70 open agricultural datasets were identified – mostly national-level statistics, not farm-level data. The majority provide only aggregate figures, not the granularity needed for AI applications (Briter, 2026).
- The cost of IoT devices, laboratory soil analysis, and satellite imagery at sufficient resolution remains prohibitive for most early-stage startups. Storage costs add a further layer that is often overlooked in early budgets.
- Field-based collection dominates (field agents, sensors, drones), making agricultural data highly terrain-dependent – each new geography restarts the data collection cycle.
- Regulatory constraints further complicate collection: drone flight authorisations, sensor deployment permits, and data capture regulations vary across jurisdictions.
- Many startups would prefer to purchase ready-made datasets, but this market is virtually non-existent in African agriculture – forcing proprietary collection even when startups lack the resources for it.
- The high cost of collection creates a perverse dynamic: startups that invest heavily in data treat it as a competitive moat, reinforcing data silos and directly limiting sharing across the ecosystem.

During hackathon preparation, several startups were unable to participate due to insufficient or unusable data – confirming that data readiness is a binding precondition for AI innovation. Rhea Soil's challenge was explicitly designed to reduce the cost of physical soil analysis by using geospatial proxy data. Apollo Agriculture built its credit model through a "lend to learn" strategy – extending credit to accumulate the repayment data needed for scoring.

IMPLICATION

- Public investment in granular, regularly updated agricultural datasets would lower the barrier to entry for startups that currently cannot afford proprietary collection.
- National strategies for geospatial data production – supported by public IoT sensor networks and reduced equipment costs – would create foundational data infrastructure directly exploitable by startups (Data 354, 2026).

CHECK USE CASE 1: APOLLO AGRICULTURE

Apollo's "lend to learn" strategy – extending credit to accumulate repayment data – shows how startups build proprietary datasets through product design when public data is unavailable.

•↔• C. Sharing data: *Data sharing is constrained as much by incentives and trust as by regulation and technology.*

Data sharing challenges affect all stakeholders across the agricultural data ecosystem – public agencies rarely share across ministries, telecoms and banks have limited incentives to open their datasets, and international actors often define the terms of data exchange. However, startups are particularly impacted: they lack the legal expertise to navigate complex sharing frameworks, the negotiating power to secure favourable terms, and the financial resources to invest in anonymisation and compliance infrastructure.

- Public agencies collect agricultural statistics but rarely share them with private sector innovators. Cross-border data sharing remains constrained by divergent data localisation laws across African markets.
- Startups face a double bind: they need access to complementary datasets (telecom data for credit scoring, land registries for traceability, weather data for insurance) but are reluctant to share their own hard-won proprietary data.
- Investors reinforce this dynamic – data is increasingly viewed as a core asset, and sharing it is perceived as eroding competitive advantage. As one investor noted in the Briter study: "data is like gold in Africa."
- A critical but under-discussed distinction exists between one-off dataset transfers and continuous data flows via APIs – the latter requiring significantly more technical architecture, trust, and ongoing governance.

- The Tolbi hackathon demonstrated that sharing can work under specific conditions: a partnership between the startup (Tolbi) and a national technical institution (BNEDT), integrated in the AI4SU program, with an open-source output framework and clear mutual benefit. The result – open-source crop classification models that neither actor could have built alone – shows what structured collaboration can produce when enabling conditions are met.

IMPLICATION

- Data sharing requires trust infrastructure: standardised data-sharing contracts, clear licensing terms, and credible intermediaries that reduce perceived risk for all parties. The development of ready-to-use contractual templates and data trust frameworks – as recommended in the Data Without Borders consultations – would replace costly case-by-case negotiations with institutionalised mechanisms (Data 354, 2026).
- Innovation funding conditioned on data openness and sharing would create economic incentives for startups to contribute to shared data commons while financing their own development (Data 354, 2026).

CHECK USE CASE 8: TOLBI

a partnership between Tolbi and the BNEDT in Côte d'Ivoire, in the context of the AI4SU program, produced open-source crop classification models that neither actor could have built alone.



D. Exploiting data: *Even when data is accessible, the lack of capabilities to process and operationalise it prevents startups from creating scalable products.*

The gap between data access and value creation is a systemic challenge across African data ecosystems – infrastructure, talent, and tooling remain underdeveloped relative to the ambition of AI-driven agriculture. For startups, this gap is acute: they compete for scarce talent with better-funded sectors, operate with limited computational resources, and often discover that available datasets are not exploitable without significant additional investment.

- One-third of all AgTechs across the 8 study countries have adopted data and AI capabilities, but the report distinguishes between high-adoption (data or AI as core product) and mid-adoption (data or AI as enabler). The majority of mid-adoption AgTechs rely on heuristics or rule-based systems rather than trained machine learning models (Briter, 2026).
- The critical talent gap is not just in data science but in hybrid profiles combining data/AI skills with agricultural domain expertise. This shortage is compounded by a structural geography mismatch: AI talent concentrates in urban centres, while agricultural problems – and the data they generate – are rural. Few training institutions offer agriculture-specific AI curricula, and those that do struggle to compete with fintech and international employers for graduates (Briter, 2026).
- Compute and storage costs are a material barrier: 22% of all funding deals for data and AI-driven AgTechs are under \$50,000 in ticket size (Briter, 2026), reflecting the early-stage nature of most ventures and limited resources available for infrastructure investment.

The exploitation gap is particularly visible in the relationship between open data and actual usability. Publicly available datasets — including European geospatial data, global soil databases, or open weather APIs — are often technically inaccessible to African startups: wrong spatial resolution for plot-level analysis, undocumented formats, or missing contextual metadata that makes the data meaningless without local ground-truthing. During the hackathons, participants repeatedly discovered that available datasets required substantial preprocessing and local calibration before any model training could begin.

Open data does not mean exploitable data — and the cost of bridging this gap falls disproportionately on startups with the fewest resources.

- Many promising use cases remain stuck at the ideation or POC stage, unable to bridge the gap to deployable products due to insufficient data volume for model training and absent ground-truth feedback loops for validation.

Across the hackathons, participants often achieved strong results with relatively simple models — confirming that in data-scarce environments, **feature engineering and domain knowledge matter more than algorithmic sophistication**.

IMPLICATION

- Talent development programs should target agriculture-specific AI skills and hybrid profiles — not generic data science curricula disconnected from the sector's realities.
- Pooling existing computing infrastructure — through shared access agreements between research institutions and startups — would give early-stage companies access to the processing capacity they need without prohibitive investment. Regional data training programmes could complement this by building the talent pipeline at scale (Data 354, 2026).

CHECK USE CASE 3: RHEA SOIL

participants used geospatial proxy data to predict soil health without laboratory analysis — confirming that domain knowledge and feature engineering can compensate for data scarcity.



E. Governing data: *Regulatory uncertainty, more than regulation itself, is what constrains startup growth.*

Data governance frameworks across the eight study countries are diverse and evolving — from Ghana and Kenya's focus on right to explanation, to Tanzania and Rwanda's data localisation requirements, to Côte d'Ivoire and South Africa's DPO registration mandates (Briter, 2026). This heterogeneity creates a complex environment for all actors. For startups, the challenge is not an excess of regulation but a deficit of clarity: rules that are difficult to find, understand, and operationalise — compounded by a near-total absence of regulatory pedagogy targeting early-stage companies.

- Startups do not reject governance — they want comprehensible and applicable rules. The problem is that most regulatory frameworks were designed for large institutions, not for resource-constrained companies iterating rapidly.

- The compliance vs. speed dilemma is acute: startups must choose between investing in regulatory conformity (slowing down) or moving fast and risking non-compliance – a choice that established players rarely face.
- Cross-border scaling multiplies the burden: a startup operating across Kenya, Tanzania, and Rwanda must navigate three different regulatory frameworks, with divergent requirements on localisation, consent, and data processing.
- Regional harmonisation through the African Union Data Policy Framework (AUDPF) is under development, but implementation remains at the framework stage – making continental harmonisation an unrealistic short-term expectation for startups planning their next 12–18 months.
- Regulatory activity is increasing: Kenya's ODPC published over 150 determinations in 2025, up from 118 in 2024, including fines for non-compliance with data protection laws (Briter, 2026). Rwanda has pioneered an AI regulatory sandbox under its National AI Policy (2023), with AgTechs among the first participants.
- The question of data sovereignty is particularly salient in agricultural value chains shaped by international compliance requirements. The EU Deforestation Regulation (EUDR) illustrates the dynamic: traceability data flows are structured to meet importing market requirements, raising the question of whether the resulting data infrastructure serves African producer interests or primarily benefits external compliance systems. When compliance datasets are hosted, processed, and governed outside the producing countries, the value generated from this data – risk analytics, supply chain optimisation, market access intelligence – accrues elsewhere. This argument is developed further in the traceability use case analysis (UC8).

During hackathon design, governance questions arose repeatedly: structuring data-sharing agreements between startups and participants required navigating IP protection, anonymisation standards, and cross-border data transfer constraints – confirming that governance is not an abstract policy concern but an operational barrier encountered at every stage. The Data Without Borders consultations (Abidjan, February 2026) reinforced this finding: without clear rules, actors default to non-sharing; without structured data, compliance verification is impossible (Data 354 / Digital Africa, 2026).

IMPLICATION

- Governance frameworks must balance protection with innovation – and regulatory pedagogy targeting startups specifically would do more to improve compliance than additional enforcement alone.
- Sector-specific governance guides and pooled compliance support – such as shared legal counsel across accelerator cohorts – would give startups the tools to integrate sound data practices from their earliest stages without disproportionate resource investment (Data 354, 2026).
- AI regulatory sandboxes (as pioneered by Rwanda) offer a practical mechanism for testing governance approaches without premature lock-in.

CHECK USE CASE 8: EUDR

the sovereignty dimension of governance: compliance data flows structured to meet EU importing market requirements raise the question of whether the resulting infrastructure serves African producer interests or primarily benefits external compliance systems.

III. USE CASES : AI & DATA IN PRACTICE

The nine use cases analyzed in this section were identified and refined through a structured co-design process conducted with more than sixty agritech startups across eight African markets. Each use case was selected on the basis of its relevance to real startup operations, its connection to concrete data and AI challenges, and its potential for operational impact beyond the proof-of-concept stage.

They span four thematic clusters:

1. **Finance & Risk** (access to finance, agricultural insurance),
 2. **Markets & Supply Chains** (market information, supply chain efficiency),
 3. **Productivity & Sustainability** (agronomic advisory, input optimization, crop risk detection),
 4. and **Compliance & Market Access** (traceability and regulated markets)
- plus one cross-cutting enabler addressing the accessibility conditions that determine whether any of the above solutions actually reaches the farmers who need them.

Each use case is structured around a consistent analytical template:

- the problem being addressed,
- the role of AI and data in the solution,
- illustrative startup approaches drawn from the ecosystem,
- data challenges specific to that context,
- market and scaling dynamics,
- risks and limitations,
- data governance considerations,
- implementation requirements,
- and the AI4SU hackathon connection where relevant.

Use cases vary in their AI intensity — some rely on genuinely central machine learning techniques, others on heuristics, domain expertise, and operational execution — and this distinction is made explicit throughout, rather than assumed.

LIST OF USE CASES

- | | |
|--|---|
| UC1 Supporting access to finance | UC2 Climate risk & insurance |
| UC3 Market information & price transparency | UC4 Supply chain & market access |
| UC5 Agronomic advisory & productivity | UC6 Input optimization & sustainable farming |
| UC7 Early crop risk detection | UC8 Traceability & regulated market access |
| UC9 AI accessibility & farmer interfaces | |

Use Case N°1

AgriTech

FinTech

Supporting smallholder farmers' access to finance

WHO BENEFITS Smallholder farmers, financial institutions (banks, MFIs, fintechs), agritech startups.

AI4SU PARTNER FarmSawa (Kenya)

VC ATTRACTIVENESS Embedded models win.



Why it matters now

Financial exclusion remains a structural barrier in African agriculture. Smallholder farmers – who represent an estimated 90% of farmers in sub-Saharan Africa – overwhelmingly lack access to formal credit¹, while the unmet financing gap for smallholder farmers across Africa, Latin America and Asia is estimated at \$170 billion². The expansion of mobile money and digital farmer platforms is creating new data trails that can support credit assessment where traditional financial records do not exist.

KEY TAKEAWAYS

- Alternative data can improve credit assessment where formal financial records are absent – but data sufficiency for a pilot does not translate into a scalable lending model.
- In thin-file rural markets, scoring models depend as much on distribution channels and operational discipline as on algorithmic performance.
- Standalone scoring is a development tool; integrated scoring is a business. The most fundable models embed credit scoring within a broader commercial offering – input finance, platform services, or lender enablement.

1. Problem addressed

Access to finance is one of the most persistent barriers facing smallholder farmers across Africa. Traditional credit systems require formal financial histories, documented income, and collateral – criteria that most smallholder farmers cannot meet. This reflects a structural mismatch between the operating realities of smallholder farming (seasonality, informality, small transaction sizes, geographic dispersion) and the risk frameworks designed for formal economic actors.

2. AI / Data approach

Startups build alternative credit scoring models using indirect signals rather than formal financial histories. A minimum viable dataset combines farmer profiles, transaction records, and early repayment behavior. More advanced approaches add satellite imagery, weather data, and agronomic indicators as productivity and climate-risk proxies.

The AI contribution here is best understood as structured risk heuristics: in thin-file contexts, these systems function less as fully predictive ML models than as operationalized combinations of proxy-based scoring, underwriting rules, and field agent judgment. A genuine ML-based scoring system requires labeled repayment data at scale – data that takes years to accumulate.

1. Savoy, C.M. (2022). "Access to Finance for Smallholder Farmers." Center for Strategic and International Studies (CSIS), December 7, 2022. <https://www.csis.org/analysis/access-finance-smallholder-farmers>

2. Ibid. Original estimate from Mastercard Foundation / ISF Advisors, cited in Savoy (2022). Note: the \$170 billion figure covers Africa, Latin America and Asia combined – not Africa alone.

Early-stage startups are almost always operating in the heuristics phase, not the ML phase, regardless of how the product is described. Recognizing this distinction is important for realistic program and investment expectations.

A structural constraint: data sufficient for an MVP does not produce the repayment history needed to generalize to new geographies or crop systems. Each new market effectively restarts the data accumulation cycle.

3. Startup approaches

- **Apollo Agriculture** (Kenya) is among the most documented integrated agri-finance models on the continent³. It combines farm-level data, satellite imagery, and repayment behavior to support lending bundled with inputs, insurance, and advisory. Its credit model was built through a 'lend to learn' strategy – extending credit to a diverse farmer base to accumulate repayment data before refining scoring algorithms. Apollo has served over 350,000 smallholder farmers in Kenya and Zambia.
- **FarmSawa** (Kenya) focuses on inclusive credit scoring for farmers with limited or no formal credit history, using alternative data signals including mobile and behavioral data to connect farmers to lenders. FarmSawa was the AI4SU program partner for the credit scoring hackathon (see Section 9).
- **Farmerline** (Ghana) illustrates how farmer data infrastructure built for advisory and platform services can be extended to support financial profiling⁴. Through its Mergdata platform, active in Ghana and Côte d'Ivoire and supporting 2.3 million farmers, Farmerline enables lender partnerships by aggregating farmer-level data at scale.

These examples show that 'credit scoring' is not a single model archetype. It can be embedded in input distribution, anchored in platform data, or integrated into financial service bundles – with meaningfully different margin profiles, risk structures, and scaling trajectories in each configuration.

4. Data challenges

Relevant data rarely exists in directly usable form. Startups must assemble farmer profiles from fragmented records, field-collected data, and digital transactions. The most valuable datasets – telecom and banking data – are controlled by third parties whose willingness to share depends on contractual leverage as much as technical feasibility. Data distributed across telecoms, agribusinesses, cooperatives, and public institutions has limited interoperability and uneven sharing incentives.

Datasets sufficient for a pilot are frequently insufficient for cross-market expansion. Generalizing a model to a new geography or crop system requires repayment history from that context – history that does not yet exist. This is less a technical problem than a structural one: scaling is slower and more capital-intensive than it appears at pilot stage.

5. Market and scaling dynamics

Alternative credit scoring benefits from mobile money infrastructure maturity, growing digital lending markets, and active DFI interest in agricultural finance – making fundraising conditions more favorable than for hardware-heavy agritech models. The most credible venture cases are integrated businesses where scoring supports a broader commercial position. Investors consistently back the business model surrounding the scoring, not the scoring itself.

3. Apollo Agriculture. Company website: <https://www.apolloagriculture.com>. See also: Shu, C. (2022). "Kenya-based agritech Apollo raises \$40 million in SoftBank-led round." TechCrunch, March 21, 2022.

4. Farmerline. Company website: <https://farmerline.co>. Mergdata platform documentation.

6. Risks and limitations

- Model reliability remains limited where data is incomplete, biased, or highly volatile.
- Dependency on third-party data providers constrains control over key inputs and margins.
- Correlated shocks (climate events, price collapses) can simultaneously undermine portfolio performance across many borrowers.
- Farmer trust is fragile when credit decisions are automated and poorly explained.
- Model transferability across geographies and crop systems is structurally limited; local recalibration is required at each expansion step.

7. Data governance and compliance

Key governance concerns in low-literacy, low-digital-agency contexts: informed consent and data reuse (transactional data collected for advisory repurposed for lending decisions); explainability of automated credit decisions; bias risks from geographically or demographically narrow training data; and function creep in integrated platform models where data flows are not transparently disclosed. Data sharing across multiple actors requires clear contractual arrangements on ownership, permitted uses, and retention – frameworks rarely standardized across these markets.

8. Implementation considerations

Viable deployment requires labeled repayment data for model training, continuous recalibration under shocks, strong distribution and collections channels, and durable institutional partnerships. Execution discipline and embedded distribution consistently matter more than scoring sophistication. The most operationally resilient models treat credit scoring as a component of a broader system, not as the system itself.

AI4SU PROGRAM CONNECTION

The AI4SU program engaged with this use case through the **Inclusive Credit Scoring for Smallholder Farmers hackathon**, organized with FarmSawa in Nairobi, Kenya (October 2025).

The challenge asked participants to develop data-driven approaches to assess creditworthiness for farmers with limited or no formal credit history.

3 AI models were produced:

- a hybrid scoring index combining satellite imagery and mobile data signals;
- a behavior-based predictive model estimating repayment likelihood;
- and an explainable AI (XAI) risk dashboard embedding fairness indicators – directly addressing the governance and explainability challenges identified in Section 7.

Use Case N°2

AgriTech

Insurtech

Climatech

Protecting farmers against climate shocks and agricultural risks

WHO BENEFITS Smallholder farmers, insurers, reinsurers, governments, startups, financial institutions

AI4SU PARTNER FarmSawa (Kenya)

VC ATTRACTIVENESS Strategically important, commercially difficult.



Why it matters now

In 2023, an estimated 97% of African smallholder farmers remained without agricultural insurance coverage⁵, despite 95% of Africa's arable land being rain-fed and heavily exposed to climate variability⁶. Insurance penetration across all segments in Africa remains below 3% of GDP⁷. Growing availability of satellite and weather data is enabling index-based products that were previously operationally impossible to deploy at smallholder scale.

KEY TAKEAWAYS

- Index-based insurance enables coverage where traditional loss verification is not operationally viable – but basis risk (payouts misaligned with actual losses) is the primary trust and adoption barrier, not price.
- Insurance scales most credibly as a risk-mitigation layer within a broader platform – agri-lending, input financing, value chain services – rather than as a standalone product.
- For program and policy design: scaling agricultural insurance requires distribution infrastructure and risk-sharing mechanisms first, data modeling second.

1. Problem addressed

Smallholder farmers face multiple risks – climate shocks, pest outbreaks, yield variability – amplified by limited irrigation, high rainfall dependency, fragmented markets, and weak public safety nets. Traditional insurance is not viable at this scale: loss verification is prohibitively costly for geographically dispersed smallholdings, and reliable ground-truth yield data is largely absent. Most farmers remain uninsured and without the risk buffer needed to invest in productivity improvements.

2. AI / Data approach

Startups and insurers use alternative data to model agricultural risk and enable scalable products without individual loss assessment. A minimum viable approach uses weather data (rainfall, temperature thresholds) and farmer location. More advanced approaches integrate satellite vegetation indices, historical climate patterns, and area-yield estimates.

5. ACRE Africa / Financial Protection Forum: "Only 3% of smallholder farmers in Africa have access to agricultural insurance coverage." See also: Africa Insurance Pulse 2023, Faber Consulting / Swiss Re. Note: the "97%" figure is derived (100% – 3%)

6. Stockholm International Water Institute (SIWI). "Transforming Investments in African Rainfed Agriculture." <https://siwi.org/publications/transforming-investments-in-african-rainfed-agriculture/>

7. Swiss Re, sigma – World Insurance reports. See also: Alhassan, A.L. & Fiador, V. (2014). "Institutional determinants of insurance penetration in Africa." Geneva Papers on Risk and Insurance, reporting Africa penetration at 2.8% in 2019 (excluding South Africa, penetration is significantly lower)

Three index-based model types coexist: weather index insurance (payout triggered by predefined thresholds); area-yield index insurance (based on zone-level average yields); and hybrid models combining index triggers with limited field validation. AI contributes to trigger design, calibration, basis-risk mapping, and anomaly detection. The critical insight: model sophistication is secondary to calibration quality. A poorly calibrated simple model generates more basis risk than a well-calibrated one – undermining adoption regardless of technical elegance.

3. Startup approaches

- **Pula** (Kenya) works with insurers and governments to design and distribute index-based insurance products across East and West Africa, typically bundled with inputs or credit. Pula positions itself as a technical and distribution enabler for insurers rather than a direct underwriter – giving it flexibility across regulatory environments.
- **ACRE Africa** (Kenya) focuses on weather-index insurance distributed through agricultural value chains, reaching over 400,000 smallholder farmers cumulatively across East and Central Africa⁸ – demonstrating the viability of value-chain distribution as the primary scaling mechanism.
- **OKO** (West Africa – Mali, Côte d'Ivoire, Uganda) develops digital crop insurance using satellite and weather data to automate payouts. Its more direct farmer-facing distribution model tests the limits of affordability and willingness to pay without bundling.

The contrast between these approaches is analytically useful: Pula and ACRE Africa scale through B2B partnerships and bundling, which improves distribution economics but creates intermediary dependence. OKO's more direct model improves farmer connection but faces steeper monetization constraints.

4. Data challenges

Ground-truth yield and loss data is sparse, inconsistently collected, and difficult to validate – making model calibration an ongoing process rather than a one-time design step. Basis risk emerges directly from this gap: when the index signal diverges from actual farm-level loss, farmers who experience losses without payouts lose trust rapidly and do not renew. Systemic climate risks create correlated losses structurally difficult to manage at scale without reinsurance.

5. Market and scaling dynamics

Agricultural insurance is strategically important but commercially fragile as a standalone product. Low margins, volume requirements, and strong reinsurance dependence make pure-play models financially vulnerable. The most credible venture-backed approaches position insurance as a risk-mitigation layer within broader platforms – where it improves portfolio performance, reduces defaults, and strengthens customer retention, rather than being the primary revenue source.

In many markets, blended finance structures or public risk-sharing mechanisms are necessary conditions for financial sustainability. Cross-country scaling is constrained by regulatory differences, climate pattern variation, and the need to rebuild distribution ecosystems in each market.

8. ACRE Africa. Company website: <https://acreafrica.com/about-us/>. See also: Kariuki, R. (2017). "Lessons Learned from ACRE Africa." Presentation, Swiss Agency for Development and Cooperation (SDC). In 2016, over 415,000 farmers were insured cumulatively since 2009.

6. Risks and limitations

- Basis risk leading to trust collapse and non-renewal if payouts do not match perceived losses.
- Correlated climate shocks creating solvency pressure and strong reinsurance dependence.
- High onboarding, farmer education, and renewal servicing costs.
- Regulatory and consumer protection risks around payout transparency and dispute resolution.

7. Data governance and compliance

Key considerations: transparency of payout mechanisms and trigger definitions communicated to farmers; fairness and geographic bias risks in model design; data-sharing arrangements between startups, insurers, reinsurers, and public partners; and conditions under which farmer data collected for insurance is used for other functions (credit, advisory). Farmer understanding of product mechanics is a governance requirement, not just a marketing challenge.

8. Implementation considerations

Successful deployment requires continuous model calibration from real-world data, strong distribution channels, clear farmer communication of product mechanics, and aligned incentives across all stakeholders. Success depends as much on ecosystem trust – earned through reliable payouts and transparent communication – as on technical model quality.

AI4SU PROGRAM CONNECTION

The AI4SU program engaged with this use case through the **Climate & Market Resilience hackathon**, organized with FarmSawa in Nairobi, Kenya (December 2025). Teams built solutions to anticipate climate shocks and market volatility to protect smallholder farmer incomes. Three models were produced, combining climate risk signals with market intelligence – directly addressing the calibration and basis-risk challenges identified in Section 4.

Use Case N°3

AgriTech

Market Intelligence

Helping farmers sell better through market information

WHO BENEFITS

Smallholder farmers, aggregators, traders, agritech startups, off-takers

AI4SU PARTNER

agriBORA (Kenya), AfriMash (Ghana)

VC ATTRACTIVENESS

Information alone doesn't monetize



Note: This use case focuses on data and information services that help farmers and value chain actors make better selling decisions. It is analytically distinct from UC4, which addresses operational and logistical execution. Startups that operate across both layers are noted explicitly; the distinction is maintained because the two involve different capital profiles, margin structures, and AI contributions.

Why it matters now

Price information in African agricultural markets remains fragmented, opaque, and highly localized⁹. Many smallholder farmers sell based on the first available buyer or immediate liquidity pressure, creating systematic information asymmetries that reduce bargaining power. Digitization is creating new opportunities to collect and distribute market data at scale – but translating information access into better farmer outcomes requires more than a price feed.

KEY TAKEAWAYS

- Price information asymmetry is a real constraint – but information alone rarely changes farmer outcomes without accompanying market access mechanisms and liquidity.
- Monetization of standalone information services is structurally difficult; the more robust models embed market intelligence within transaction or logistics platforms where value is captured on execution.
- The systemic value of these platforms: they progressively formalize market data that was previously informal, improving coordination across value chains beyond individual transactions.

1. Problem addressed

Smallholder farmers often lack visibility on current market prices, selling to the first available buyer based on informal benchmarks. This creates systematic asymmetries between farmers and intermediaries – particularly traders operating across multiple markets with full price visibility. Price volatility compounds the problem, making optimal selling strategies difficult to anticipate without reliable, timely data.

2. AI / Data approach

A minimum viable approach relies on market price data and contextual variables (location, crop, seasonality) for descriptive analytics and trend modeling.

⁹. Descriptive characterization consistent with: World Bank (2017). "ICT in Agriculture: Connecting Smallholders to Knowledge, Networks, and Institutions." Updated ICT in Agriculture Sourcebook. See also: FAO Agricultural Market Information Systems (AMIS)

More advanced approaches apply machine learning for price modeling under sparse data (time-series and regression models), market structure learning (clustering similar markets to transfer signals), and quality control (anomaly detection, actionable selling recommendations).

The AI contribution is primarily in handling data sparsity and heterogeneity, not in generating precise forecasts. Agricultural prices are influenced by partially observable variables – informal trade, local shocks, negotiation practices – making purely predictive models fragile. Most startups complement quantitative models with rule-based systems or advisory layers, aiming to support better decisions under uncertainty rather than accurate point predictions.

3. Startup approaches

- **AgroCenta** (Ghana) connects farmers directly to buyers while providing market price information, reducing reliance on intermediaries in smallholder grain and commodity markets.
- **agriBORA** (Kenya) uses market intelligence to support farmer selling decisions within a broader model combining warehousing, credit, and market linkage. In this use case, agriBORA's relevant role is the information and decision-support layer: price signals, market analysis, and selling recommendations that help farmers decide when and where to sell. Its logistics and execution operations are addressed in UC4.
- **Twiga Foods** (Kenya) generates and uses market data as an input to its pricing and distribution decisions. In this use case, the relevant dimension is Twiga's contribution to price signal formation and market transparency across its aggregation network. Its logistics and supply chain operations are addressed in UC4.

A consistent pattern: the most impactful market information services are not standalone price feeds but embedded components of platforms that also provide market access, financing, or logistics. Progressive formalization of previously informal market data creates systemic coordination value beyond individual commercial transactions.

4. Data challenges

Market data is structurally difficult to access: traders and intermediaries may have limited incentives to share price information, as opacity is a competitive advantage. Price is not a single ground-truth value – it is negotiated and depends on quality grade, volume, units, timing, and transport, making standardization a prerequisite for reliable modeling. Farmer-generated data presents additional challenges: irregular reporting, inconsistent units, and limited digital tools require significant cleaning before supporting inference.

5. Market and scaling dynamics

Standalone market information services face a structural monetization challenge: farmers benefit from price information but are often not willing or able to pay directly. Value capture requires integration within transaction, logistics, or bundled service platforms. Information-only models are asset-light but struggle with revenue; transaction-enabled models can capture more value but require working capital, logistics coordination, and local operational presence.

A key behavioral constraint: access to information does not guarantee changed farmer behavior. Liquidity constraints, product perishability, and restricted buyer access can prevent farmers from acting on available information even when it is accurate and timely.

6. Risks and limitations

- Low impact when solutions focus only on information without enabling market access.
- Model fragility due to high price volatility and informal market dynamics.
- Trust erosion after incorrect signals – farmers who act on wrong information and lose out disengage rapidly.
- Resistance from intermediaries who benefit from opacity.

7. Data governance and compliance

Governance challenges include: ownership of platform-collected market data; risk of concentration among dominant platform actors; transparency of recommendations and their basis; and conditions under which farmer-generated data is used for purposes beyond the original service. Public actors can support price transparency initiatives, but alignment with private sector incentives is rarely automatic.

8. Implementation considerations

Operational success requires reliable data pipelines, standardization and quality control, tools adapted to low-connectivity environments, and combination of insights with intermediation or advisory. Solutions pairing information with execution capabilities consistently outperform purely information-driven approaches.

AI4SU PROGRAM CONNECTION

The **Maize Price Intelligence for Smarter Selling hackathon** with agriBORA (November–December 2025) produced three ML models for weekly maize price forecasting across Kenyan counties – illustrating both the feasibility and the inherent uncertainty limits of price forecasting in volatile agricultural markets.

The **Customer Segmentation & Retention Intelligence hackathon** with AfriMash (Ghana, October 2025) complements this use case from the demand-intelligence angle: RFM-based churn/propensity models and LTV pipelines for agri-e-commerce platforms directly address the buyer-side data gap in market information systems.

Use Case N°4

AgriTech

Supply chain

FoodTech

Improving market access through supply chain efficiency

WHO BENEFITS

Smallholder farmers, aggregators, buyers, distributors, agritech startups, off-takers

AI4SU PARTNER

agriBORA (Kenya), AfriMash (Ghana)

VC ATTRACTIVENESS

Execution over algorithms



Note: This use case focuses on the operational and logistical execution layer of market access – aggregation, transportation, and commercialization. It is analytically distinct from UC3, which addresses market information and price signals. Both layers are necessary: knowing where and when to sell does not ensure the ability to do so efficiently. Startups that operate across both layers are noted with explicit distinction of the relevant role in each use case.

Why it matters now

Post-harvest losses in developing countries are estimated at 30–40% of total food production¹⁰ – a composite figure that varies significantly by commodity and region. In Africa, these losses are driven primarily by logistics gaps and supply-demand mismatches rather than data shortfalls. While digital market information has improved, the physical ability to move products efficiently and connect supply with demand remains a primary structural bottleneck to farmer income.

KEY TAKEAWAYS

- Value creation in this use case is primarily driven by execution capabilities – logistics, aggregation, physical distribution – not by predictive AI. Data-driven optimization improves efficiency at the margins.
- Defensibility comes from operational density and network control, not from data models. Each new market requires rebuilding supply chains and partnerships, limiting cross-country replication.
- For investment design: thin margins and capital intensity make profitability highly sensitive to execution quality and volume concentration – not to model sophistication.

1. Problem addressed

Agricultural supply chains in many African contexts are fragmented and inefficient. Farmers face challenges transporting produce to markets, accessing reliable buyers, and coordinating volumes at scale. These constraints generate high post-harvest losses, weak bargaining power, and elevated transaction costs. Supply and demand are often poorly synchronized, creating mismatches that benefit intermediaries at the expense of farmers and end buyers.

2. AI / Data approach

Startups use data to improve coordination and operational efficiency. A minimum viable approach uses farmer and production data, buyer demand data, and logistics routing information.

10. FAO (2011). "Seeking end to loss and waste of food along production chain." FAO-SAVE FOOD Initiative.

<https://www.fao.org/4/i2940e/i2940e02.pdf>. Note: this is a composite estimate across developing countries, not specific to a single African value chain.

Typical applications include supply-demand matching, logistics optimization (routing, fleet management, delivery planning), volume estimation, and operational pricing integrating logistics constraints.

In practice, most solutions use relatively simple models, heuristics, or rule-based systems. Being data-driven and operationally reliable is more critical than deploying advanced algorithms. The AI contribution is in coordination and optimization – the baseline it replaces is manual coordination, which is slow and error-prone, not a sophisticated analytic system.

3. Startup approaches

- **Twiga Foods** (Kenya) operates an asset-heavy integrated model combining aggregation, cold chain logistics, and distribution to supply retailers and vendors across Kenyan urban markets. In this use case – distinct from its market information role in UC3 – Twiga's relevant dimension is its physical supply chain: route optimization, volume management, and last-mile distribution infrastructure.
- **agriBORA** (Kenya) represents an asset-light approach focused on market linkage – structuring transactions and connecting farmers to buyers while integrating storage and financing services. In this use case – distinct from its price intelligence role in UC3 – agriBORA's relevant dimension is its operational coordination capability: linking supply to demand and managing product movement without owning physical infrastructure.
- **Complete Farmer** (Ghana) illustrates a structured B2B sourcing and fulfillment model, connecting growers and buyers through coordinated sourcing, quality control, and commercialization for export and large-buyer markets.
- **Farm to Feed** (Kenya) focuses on recovering and redistributing surplus produce, reducing food waste by connecting surplus supply to demand channels. Its recommender system – developed through the AI4SU hackathon – predicts purchase likelihood and optimal redistribution pathways, demonstrating how demand intelligence can improve supply chain efficiency from the waste-reduction angle.

4. Data challenges

Supply data is often unreliable: farmer-reported volumes are inconsistent and real-time data is difficult to obtain in environments requiring high reactivity. Demand data may be fragmented across buyers. Standardization of units, quality grades, and pricing structures is a prerequisite for optimization at scale – a requirement rarely met at the start of operations and built iteratively through operational experience.

5. Market and scaling dynamics

Monetization pathways include transaction fees, margins on traded goods, and bundled services. However, these models combine thin margins with high logistics costs, inventory risk, and working capital constraints – making profitability highly sensitive to execution quality and volume concentration. Asset-heavy models generate stronger operational defensibility but require significant upfront capital. Asset-light models deploy faster but may struggle to capture durable value without complementary services.

Defensibility emerges from operational density, supply and demand network control, and long-term buyer relationships. Each new market requires rebuilding these elements, constraining cross-country replication and limiting the transfer value of data models built in one context.

6. Risks and limitations

- High operational complexity and execution risk.
- Low margins and strong volume dependence.
- Capital intensity for logistics infrastructure and inventory financing.
- Difficulty ensuring consistent supply quality and reliability at scale.
- Replication of similar models across geographies limits long-term differentiation.

7. Data governance and compliance

Governance challenges center on ownership of transaction and supply chain data, concentration risk among dominant platform actors accumulating data advantage over time, and transparency of pricing and matching mechanisms. Ensuring fair platform data access for smaller value chain actors – cooperatives, smallholder aggregators – is an important consideration as these models scale.

8. Implementation considerations

Successful deployment requires reliable supply networks, consistent data collection and standardization, efficient logistics coordination, and sustained trust with farmers and buyers. The primary challenge is not generating insights but translating them into effective, scalable operations – a fundamentally operational and relational challenge, not a technical one.

AI4SU PROGRAM CONNECTION

The **Produce Recommendation Systems for Food Waste Reduction hackathon** with Farm to Feed (November–December 2025) produced three recommender models using anonymized transaction data to predict purchase likelihood (7 and 14 days ahead) per customer and SKU – illustrating how demand-side intelligence can improve supply chain coordination and reduce waste simultaneously.

The **Customer Segmentation & Retention Intelligence hackathon** with AfriMash (Ghana, October 2025) complements this use case: RFM-based churn/propensity models and LTV pipelines improve how agri-e-commerce platforms match supply with buyer demand – a practical complement to the logistics optimization layer.

Use Case N°5

AgriTech

Advisory

Climatech

Supporting farmers' agronomic decisions and productivity

WHO BENEFITS

Smallholder farmers, agribusinesses, input providers, agritech startups.

AI4SU PARTNER

DigiCow (Kenya)

VC ATTRACTIVENESS

Easy to build, hard to monetize



Note: This use case addresses multi-variable decision support across the full production cycle – planting, crop management, risk anticipation, and market timing. It is analytically distinct from UC6, which focuses on input use optimization specifically, with a direct commercial link to input supply chains and sustainability compliance requirements. UC5 is about integrated agronomic judgment across multiple farm variables; UC6 is about input efficiency and its economic and environmental consequences.

Why it matters now

Improving yields and resilience is a central challenge in African agriculture, under growing climate variability and soil degradation pressure. Smallholder farmers operate in highly variable environments with limited access to agronomic expertise. Increasing availability of weather, satellite, and agronomic data is enabling new decision-support systems – but their real-world impact is more conditional on adoption and distribution than their technical potential suggests.

KEY TAKEAWAYS

- Multi-variable decision support requires combining heterogeneous data sources; single-factor optimization delivers limited impact in complex farming environments.
- Advisory models are among the easiest to build and the hardest to monetize directly – willingness to pay is low, and value is captured primarily by input suppliers, lenders, and agribusinesses upstream.
- The adoption constraint – farmers not acting on available recommendations – is as important as the data quality constraint. Predicting and targeting adoption propensity is an actionable and underexplored intervention for program design.

1. Problem addressed

Smallholder farmers must constantly arbitrate between multiple constraints – weather, soil conditions, input availability, market timing – without integrated decision-support tools. Decisions on planting, input application, and risk management are based on incomplete or informal information, resulting in suboptimal yields, inefficient resource use, and increased vulnerability to climate and biological risks.

2. AI / Data approach

This use case relies on combining heterogeneous data sources: weather data and basic farm information at a minimum; satellite imagery, drone data, sensor data (soil moisture), and scientific agronomic datasets at more advanced levels.

These support yield and stress estimation, multi-variable decision-support systems, and recommendation engines adapted to local conditions.

In practice, most solutions remain data-driven rather than AI-driven, relying on heuristics or simplified agronomic models. The actual AI contribution is often in pattern recognition across satellite data or in personalizing recommendations based on farmer-specific variables. Performance depends less on model sophistication than on data quality, the ability to integrate heterogeneous sources, and the availability of ground-truth feedback loops – which are rarely built into early-stage deployments.

3. Startup approaches

- **Farmerline** (Ghana) embeds multi-factor advisory within a broader farmer platform combining data, training, and market services. Its Darli AI tool, recognized on TIME's Best Inventions of 2024¹¹, illustrates how agronomic AI gains traction when embedded in a trusted platform with established farmer relationships.
- **Ignitia** (Ghana, Mali, Nigeria) provides hyper-local weather intelligence as a core agronomic decision input via SMS – demonstrating that high-impact advisory can come from a single high-quality data source well-localized, rather than complex multi-variable models.
- **DigiCow** (Kenya) provides digital advisory and practice training to smallholder dairy farmers, with a focus on improving adoption of improved farming practices. DigiCow was the AI4SU program partner for the practice adoption hackathon (see Section 9).
- **Rhea Soil** (Kenya) offers soil diagnostics as a data input to multi-variable advisory systems. Its direct input optimization application is addressed in UC6.

4. Data challenges

Combining multiple heterogeneous data sources with different formats, resolutions, and reliability levels is the core technical challenge. Satellite data is partially open but resolution-constrained; higher-resolution data is costly. The most critical bottleneck is the absence of reliable ground-truth feedback loops – farmers do not systematically report outcomes, making model validation and improvement structurally difficult.

5. Market and scaling dynamics

Advisory services are rarely monetized directly. Economic value is more consistently captured by input providers, lenders, insurers, and agribusinesses who benefit from improved farm performance. Most viable models embed advisory within broader commercial offerings. Measuring impact is structurally difficult: productivity improvements are influenced by many external factors, making it hard to attribute yield gains to advisory specifically – limiting both investor confidence and program evaluation.

11. TIME (2024). "The Best Inventions of 2024: Farmerline Darli AI." October 30, 2024. <https://time.com/collections/best-inventions-2024/7094874/farmerline-darli-ai/>

6. Risks and limitations

- Limited impact if recommendations are not adopted – adoption rates are often lower than technical reach figures suggest.
- Over-reliance on incomplete or low-quality data producing unreliable recommendations that erode trust.
- Difficulty measuring and attributing productivity impact to advisory specifically.
- Risk of overpromising AI capabilities in contexts where local agronomic knowledge and farmer relationships are the real value drivers.

7. Data governance and compliance

Key considerations: ownership and use of farm-level data collected through advisory platforms; transparency of recommendations when linked to commercial products (see UC6 for the conflict-of-interest dimension); dependency on external satellite data providers; and potential linkage to compliance-relevant practice records, connecting UC5 governance questions to the traceability issues addressed in UC8.

8. Implementation considerations

Successful deployment requires reliable integration of multiple data sources, contextualized and actionable outputs, delivery channels accessible to farmers with limited connectivity, and feedback loops for continuous improvement. The primary challenge is not generating recommendations but supporting effective decision-making in real-world farming conditions – which means addressing the last-mile delivery problem, not the modeling problem.

AI4SU PROGRAM CONNECTION

The **Predicting Practice Adoption in Smallholder Farming** hackathon with DigiCow (January–February 2026) asked participants to predict whether farmers would adopt improved practices after training – enabling smarter targeting of follow-up support and more efficient extension program design.

Three models were produced: structured preprocessing for generalization; end-to-end feature engineering and classification; and robust probability estimation.

These outputs directly address the adoption constraint flagged in Section 6: the most accurate recommendations still fail if farmers do not act on them. Predicting adoption propensity is a practical, actionable pathway to improving advisory program efficiency – and a contribution the AI4SU hackathon approach is uniquely positioned to deliver.

Use Case N°6

AgriTech

Input supply

Sustainability

Optimizing input use and supporting sustainable farming

WHO BENEFITS

Smallholder farmers, input providers, agribusinesses, buyers, public actors

AI4SU PARTNER

Rhea Soil (Kenya)

VC ATTRACTIVENESS

Clearer economic logic than advisory



Note: This use case is analytically distinct from UC5 in two ways: it focuses on input use optimization specifically – fertilizers, water, crop treatments – rather than multi-variable agronomic decision support; and it has a direct commercial link to input supply chains and to evolving sustainability compliance requirements from buyers and regulators. These two dimensions – commercial embeddedness and compliance pressure – define the strategic logic of this use case and differentiate it from general advisory.

Why it matters now

Rising input costs, soil degradation, and water stress are increasing pressure on farmers to manage resources more efficiently. At the same time, evolving export market requirements – sustainability standards, residue limits, environmental compliance obligations – are pushing agricultural systems toward more accountable input use. This creates demand for tools that support both productivity optimization and compliance demonstration, often simultaneously.

KEY TAKEAWAYS

- Input optimization has a clearer economic logic than general advisory – the benefit is measurable in reduced costs and improved yields – but adoption is constrained by affordability and behavioral inertia.
- The commercial embeddedness of input optimization creates a structural conflict of interest: advisory systems linked to input distribution may optimize for sales rather than farm-level efficiency.
- Sustainability compliance is becoming a commercial driver for input optimization, not just an environmental objective – particularly for export-oriented value chains subject to buyer or regulatory requirements.

1. Problem addressed

Farmers must manage inputs under simultaneous constraints: cost, local availability, soil variability, and climate. Inefficient input use drives high production costs, soil depletion, and chemical overuse. The challenge is not simply applying less – it is applying the right inputs, in the right quantities, at the right time, in the right locations. That requires localized, granular data that is rarely available.

Transitioning toward more sustainable input practices introduces additional trade-offs: reduced chemical use may affect short-term yields; switching to alternative inputs involves cost and behavioral change. These trade-offs are difficult to manage without decision-support tools – and harder still when compliance requirements from buyers or regulators evolve faster than farm practices can adapt.

2. AI / Data approach

A minimum viable approach combines soil data (where available), weather data, and crop and farm characteristics. More advanced approaches add sensor data, satellite imagery, and historical yield records to support dosage recommendations, timing optimization, and scenario-based input selection. Most solutions rely on agronomic rule sets and relatively simple models. The AI contribution is primarily in handling spatial variability – producing localized recommendations that generic agronomic rules cannot – and in integrating multiple data signals to identify highest-marginal-return interventions. The data constraint is more binding than the algorithmic constraint: granular soil data is expensive and unstandardized, making local calibration the primary bottleneck rather than model design.

3. Startup approaches

- **Rhea Soil** (Kenya) focuses on soil intelligence and localized input recommendations, treating soil data as a commercial asset that creates value for farmers, input suppliers optimizing distribution, and agribusinesses requiring raw material consistency. Rhea was the AI4SU program partner for the soil nutrient prediction hackathon (see Section 9).
- **Apollo Agriculture** (Kenya) integrates input provision, credit, and advisory within a bundled service model. In this use case – distinct from its credit scoring role in UC1 – Apollo's relevant dimension is how bundling input access with advisory creates a commercial feedback loop: better input recommendations improve repayment rates, which improve credit model performance, which support further lending.

A structural tension in this use case: when advisory systems are operated by or commercially linked to input distributors, the incentive to recommend optimal quantities from the farmer's perspective may diverge from the incentive to maximize input sales. This conflict of interest requires transparency mechanisms and independent data validation – particularly when systems are deployed at scale in low-literacy contexts where farmers cannot independently assess recommendation quality.

4. Data challenges

Soil data is costly to collect at scale: sampling, laboratory analysis, and spatial coverage requirements are significant constraints for smallholder segments. Sensor deployment remains limited. Strong plot-level variability reduces model generalizability – a model calibrated in one area may not transfer to a neighboring area with different soil profiles. Recommendations frequently rely on proxy data or interpolated estimates, limiting localization precision in heterogeneous farming environments.

5. Market and scaling dynamics

Input optimization is commercially attractive because it links directly to measurable farmer costs and crop quality. Value capture is typically concentrated upstream: input suppliers benefit from optimized sales; agribusinesses from improved supply consistency; financial actors from improved farm performance. For farmers, benefits are real but often diffuse and delayed – not always sufficient to drive adoption without complementary incentives.

The cost of generating high-quality localized data can be significant relative to the value created for smallholder segments specifically. Models primarily serving agribusiness or buyer interests through improved supply quality are more commercially viable than those targeting smallholder affordability directly – a gap between development impact potential and commercial scaling logic that program design should explicitly address.

6. Risks and limitations

- Recommendations not economically viable for smallholder farmers at current input prices.
- Structural conflict of interest when advisory is commercially linked to input distribution.
- AI-driven optimization may prioritize short-term productivity over long-term soil health if model objectives are not carefully designed.
- Dependency on incomplete or proxy data producing recommendations that are technically sound but locally inappropriate.

7. Data governance and compliance

Key governance issues: ownership and use of farm-level and soil data; transparency of recommendations when commercial relationships create conflicts of interest; and conditions under which compliance-related data (input records, practice documentation) can be accessed by buyers or certifying bodies. The intersection of commercial advisory and regulatory compliance creates both opportunity and governance risk – particularly for smallholders with limited ability to negotiate the terms on which their data is used.

8. Implementation considerations

Successful deployment requires: access to reliable and localized data; strong integration with input supply chains; alignment with farmer affordability constraints; and partnerships with public actors and research institutions for calibration and validation. The primary challenge is ensuring that recommendations are economically viable, trusted, and aligned with both farmer interests and compliance requirements – not technical soundness alone.

AI4SU PROGRAM CONNECTION

The **Soil Nutrient Prediction for Sustainable Farming hackathon** with Rhea Soil (February–March 2026) asked participants to estimate soil nutrient levels without laboratory testing, using geospatial data and optional earth-observation signals.

Three models were produced: a domain-aware ensemble; a climate-zone and village-level model; and a geospatial feature engineering approach.

These outputs directly address the data scarcity bottleneck in Section 4 – demonstrating how open-source AI can provide localized soil intelligence without the cost of physical laboratory analysis. More broadly, they illustrate how the AI4SU open innovation approach can reduce the data collection barrier that currently limits input optimization at smallholder scale – producing public goods rather than proprietary tools.

Use Case N°7

AgriTech

Remote sensing

Precision agriculture

Detecting crop risks early to protect farmer yields

WHO BENEFITS

Smallholder farmers, agribusinesses, input providers, insurers, agritech startups

AI4SU PARTNER

Tolbi (Côte d'Ivoire/Senegal)

VC ATTRACTIVENESS

Technically central but standalone is weak



Why it matters now

Limited access to agronomic expertise and the increasing availability of satellite, drone, and smartphone data are creating new opportunities for large-scale crop monitoring. Early detection of diseases, stress, and yield threats is becoming more critical as climate variability intensifies. This is one of the few agritech use cases where AI techniques – particularly computer vision and remote sensing – can play a technically central role, though impact remains conditional on what happens after detection.

KEY TAKEAWAYS

- Crop monitoring is one of the few agritech use cases where AI is technically central – but performance depends more on data availability, labeling quality, and local calibration than on algorithmic design.
- Standalone detection solutions face a structural impact gap: disease detection only improves outcomes if farmers can access an affordable, appropriate response.
- For program design: the detection-to-response pathway – available inputs, integrated advisory, cooperative infrastructure – requires as much attention as detection capability itself.

1. Problem addressed

Farmers often lack timely information on crop health. Diseases, nutrient deficiencies, and water stress may go undetected until yields are already significantly affected. Monitoring large or geographically dispersed plots without digital tools is operationally difficult. Early-stage symptoms are often visually indistinguishable from normal variation without specialized knowledge – making delayed detection and inadequate response a persistent source of preventable yield loss.

2. AI / Data approach

This use case relies heavily on AI applied to visual and environmental data. Key techniques include: computer vision models to detect diseases or anomalies from leaf or field images; remote sensing using satellite vegetation indices to monitor crop health and growth trends; and anomaly detection to identify early signs of stress at scale. The AI contribution is genuinely central here – the baseline it replaces is manual field inspection by agronomists, which is costly, slow, and not scalable.

Data sources include open and commercial satellite imagery, smartphone images from farmers or field agents, drone imagery, and agronomic reference datasets.

Model performance depends critically on image quality, locally relevant labeled datasets, and frequency and reliability of data collection – all structurally constrained in smallholder contexts. Building labeled agricultural disease datasets requires domain expertise, is expensive, and must be repeated as crops, regions, and conditions change.

3. Startup approaches

- **Aerobotics** (South Africa) uses aerial and satellite imagery to provide crop analytics primarily for commercial farms and agribusinesses – a B2B model with higher willingness-to-pay and more structured data environments than smallholder deployments. It illustrates both the technical maturity of remote sensing applications and the gap between commercial-scale and smallholder-scale feasibility.
- **Tolbi** (Côte d'Ivoire/Senegal) combines satellite-based crop monitoring with traceability and compliance tools, showing how monitoring can be embedded in broader market access and compliance services. Tolbi's approach demonstrates how monitoring creates value not just for farmers but for buyers and certifying bodies requiring production practice evidence. Tolbi was the AI4SU program partner for the crop classification hackathon (see Section 9).

Note: PlantVillage/Nuru (Penn State University) is a research and development initiative, not a commercial startup. It is referenced here for context on smallholder-facing mobile disease detection capabilities but is not representative of startup ecosystem dynamics or commercial viability.

4. Data challenges

Limited availability of high-quality, locally labeled datasets is the primary bottleneck. High variability of crops, diseases, and environmental conditions across regions; resolution constraints for open satellite data; and commercial imagery costs are all structural constraints that limit scalability. Building a labeled dataset for a new crop or region effectively restarts the data development process – there is limited transfer of labeling investment across contexts.

5. Market and scaling dynamics

As a standalone product, crop monitoring struggles to demonstrate direct economic value: detecting a disease only improves outcomes if farmers can access an appropriate and affordable response. Without integration into advisory, input distribution, or cooperative infrastructure, detection capability translates poorly into farm-level impact. Most viable approaches integrate monitoring into broader service offerings where it contributes indirectly to value creation by improving decision quality or reducing risk. Recent LLM and generative AI advances have expanded access to image-based diagnostics. In African agricultural contexts, their practical value remains limited by underrepresentation of local crops and conditions in training data, and by absence of integration with agricultural service ecosystems. These tools may complement specialized solutions in future but do not currently substitute for locally grounded calibration.

6. Risks and limitations

- Inaccurate or inconsistent diagnoses (false positives/negatives) eroding farmer trust.
- Dependence on sparse or low-quality labeled data producing unreliable models.
- Limited adoption due to technology access and connectivity barriers.
- The detection-to-response gap: detection capability without accessible response options has limited or no farm-level impact.

7. Data governance and compliance

Key issues: ownership of farmer-generated image data collected at scale; transparency of diagnostic outputs; dependency on external satellite data providers; and data standardization across systems used for both agronomic and compliance purposes. Image data collected for monitoring may reveal sensitive land-use or farm management information – raising questions about who has access and under what conditions.

8. Implementation considerations

Successful deployment requires locally relevant labeled datasets, partnerships with research institutions for validation, user-friendly tools for low-connectivity environments, and integration with service layers that translate detection into actionable interventions. The detection-to-response pathway deserves explicit design attention. Three conditions appear necessary for detection to consistently improve smallholder outcomes: (1) affordable and accessible treatment inputs available locally at the time of detection; (2) advisory integration that translates a diagnosis into a specific, locally available intervention – not a generic recommendation; and (3) cooperative or extension infrastructure that can aggregate input demand and support group purchasing when individual quantities are too small to access efficient supply channels. In the absence of these conditions, detection capability is necessary but not sufficient.

AI4SU PROGRAM CONNECTION

The **Pixel-Level Crop Classification for Precision Agriculture hackathon** with Tolbi (April–June 2025) produced three crop classification models from open-source Sentinel-2 satellite imagery to distinguish cocoa, rubber, and oil palm plantations in Côte d'Ivoire: a remote sensing deep learning pipeline; a Sentinel-2 time-series segmentation model; and a pixel classification approach.

With 866 participants, this was the first and most widely attended AI4SU hackathon. These outputs demonstrate the feasibility of scalable, open-source crop mapping as a foundation for both monitoring and compliance use cases – directly connecting the detection capabilities of UC7 with the compliance and traceability requirements of UC8.

Use Case N°8

AgriTech

Traceability

Compliance

Enabling access to international and regulated markets

WHO BENEFITS

Agribusinesses, exporters, buyers, regulators, and – indirectly – farmers

AI4SU PARTNER

Tolbi (Côte d'Ivoire/Senegal)

VC ATTRACTIVENESS

Regulation-driven demand is reliable



Why it matters now

The EU Deforestation Regulation (EUDR), which entered into force on 29 June 2023, now applies from 30 December 2026 for large and medium operators, and from 30 June 2027 for micro and small enterprises¹². It requires companies placing seven key agricultural commodities on the EU market to demonstrate deforestation-free sourcing. Sub-Saharan Africa risks losing up to US\$11 billion in annual export revenue – a modeled upper bound based on current trade flows – if unable to meet these requirements¹³. More than 59% of Africa's cocoa exports and 41.6% of its coffee exports are EU-bound¹⁴. In several value chains, traceability has shifted from a competitive advantage to a condition of market access.

KEY TAKEAWAYS

- Traceability is primarily driven by regulatory and buyer pressure – its strategic importance varies significantly by crop type and export exposure.
- Data serves as proof of compliance, not only as an optimization input – a fundamentally different value proposition from other use cases in this report.
- The strategic opportunity for African startups: build traceability infrastructure that generates sovereign data assets and positions African producers as data owners, not just compliance data providers for importing markets.

1. Problem addressed

Agricultural value chains in many African contexts lack reliable systems to track product origin, production practices, and compliance with international standards. Compliance processes rely on declarative or manual systems that are difficult to verify and scale. This limits access to regulated export markets, increases risk for buyers and exporters, and concentrates compliance costs on value chain actors least able to bear them – often smallholder farmers and small cooperatives who face exclusion if they cannot meet documentation requirements.

2. AI / Data approach

This use case combines geospatial data (satellite imagery for land use and deforestation monitoring), supply chain data (transaction records, aggregation point tracking, lot traceability), and producer-level data (location, declared practices).

12. Regulation (EU) 2023/1115 (entered into force 29 June 2023), as amended by Regulation (EU) 2024/3234 (December 2024) and Regulation (EU) 2025/2650 (December 2025). Current application dates: 30 Dec. 2026 (large/medium operators); 30 June 2027 (micro/small enterprises).

13. The Commonwealth Secretariat (2025). "The EU Deforestation Regulation: Trade and Investment Implications for Sub-Saharan African Countries." April 2025. <https://thecommonwealth.org/publications/eu-deforestation-regulation-trade-and-investment-implications-sub-saharan-african>. Note: the US\$11 billion figure is a modeled upper bound based on current trade flows, not an observed figure.

14. Ibid. Computation based on ITC Trade Map data (2024), average 2021–2023. Also cited in: White & Case (2025). "European Deforestation Regulation: Implications for Africa's food security." Africa Focus Autumn 2025.

AI contributes to land use classification and environmental risk detection; identification of inconsistencies or anomalies in reported data; and automation of verification processes that would be prohibitively expensive manually at smallholder scale.

The value of AI lies less in prediction than in enabling scalable verification – replacing manual controls with systematic, data-driven processes across geographically dispersed, fragmented supply chains. The credibility of these systems depends on data governance as much as on technical design: a technically sophisticated system built on unreliable inputs provides false assurance rather than genuine traceability.

3. Startup approaches

Tolbi (Côte d'Ivoire) combines satellite data and traceability tools to support compliance with environmental and export market requirements, directly addressing EUDR compliance needs for cocoa value chains in West Africa. Tolbi represents one of the most advanced African-headquartered startups building sovereign traceability infrastructure for EUDR-affected commodities. Tolbi was the AI4SU program partner for the crop classification hackathon (see Section 9).

Note: Satelligence (Netherlands) provides geospatial monitoring capabilities referenced in this domain but is not an African startup. Its inclusion in the appendix is for reference context only. The emergence of African-led alternatives to non-African compliance infrastructure is an active development – and a strategic priority for building sovereign data capacity in export-oriented African value chains.

4. Data challenges

Data is fragmented across actors and systems with limited standardization. Satellite data involves trade-offs: open data is accessible but may lack the resolution required for plot-level deforestation monitoring; commercial imagery is costly and requires significant processing capacity. Verification of field-level data remains complex and expensive to validate independently, creating vulnerability to manipulation in non-audited systems. Credibility depends less on data volume than on reliability, consistency, and governance.

5. Market and scaling dynamics

Traceability benefits from strong, regulation-driven demand – a more reliable commercial driver than most agritech use cases. Value is captured primarily by exporters and agribusinesses (market access, reduced rejection risk) and buyers (compliance and reputational risk management). Business models are B2B with comparatively higher willingness-to-pay. Farmers benefit indirectly through continued market access, but may also bear part of the compliance cost burden – a structural tension requiring careful management. Scaling is constrained by integration complexity across supply chain actors, local regulatory environments, and the need for trusted and interoperable data systems. Compliance requirements can create exclusionary dynamics: smallholders without resources to meet documentation requirements risk being pushed out of regulated export markets.

A deeper structural tension: compliance requirements are defined by importing markets, and the systems used to demonstrate compliance often generate data flows toward buyers and international platforms rather than remaining under African producer control. This defines both a risk and an opportunity.

Startups that aggregate traceability data on behalf of cooperatives or national commodity boards – rather than on behalf of buyers – can position African producers as data owners. Ghana's Cocoa Management System (CMS), developed by COCOBOD, illustrates how national-scale infrastructure can simultaneously satisfy compliance and build sovereign data assets¹⁵. Interoperable national systems, local calibration advantages, and open-source compliance tools – such as those produced through the AI4SU hackathon with Tolbi – can reduce dependency on proprietary platforms controlled by non-African actors. Supporting startups in this space should therefore go beyond technical capability to include data governance frameworks and cooperative models that position African producers as owners of their compliance infrastructure.

6. Risks and limitations

- Reliance on incomplete or unreliable data undermining the credibility of compliance claims.
- Potential manipulation or fraud in declarative reporting systems without independent verification.
- High compliance costs for smallholders and small cooperatives, risking market exclusion.
- Risk that traceability requirements accelerate supply chain consolidation, reducing smallholder market access rather than improving it.

7. Data governance and compliance

Key governance questions: who owns supply chain and farmer data generated through traceability systems; what conditions govern access by buyers, certifying bodies, and regulatory authorities; how compliance data interoperates across systems defined by different importing markets; and how African institutional frameworks can shape data governance architecture rather than simply complying with externally defined requirements. These are institutional and political questions with material consequences for how value is distributed across agricultural supply chains.

8. Implementation considerations

Successful deployment requires integration of heterogeneous data sources; partnerships with public institutions, commodity boards, and certification bodies; alignment with regulatory frameworks while maintaining flexibility for evolving requirements; and trust-building across value chain actors with different interests and compliance exposures. Startups play a critical role by providing technological agility and scalable verification where public monitoring infrastructure is limited – but their impact is maximized when embedded within institutional frameworks that maintain data sovereignty for African producers.

15. Ghana Cocoa Board (COCOBOD). "Meet-the-Press: Cocoa Management System (CMS)." October 15, 2024. <https://cocobod.gh/news/cocobod-meet-the-press>. Note: the €50M investment figure cited in the v2 draft could not be independently verified in publicly available sources; the text has been reformulated accordingly

Use Case N°9

AgriTech

Cross-cutting enabler

Digital Inclusion

Making agricultural AI and data tools accessible to all farmers

WHO BENEFITS

Smallholder farmers, agribusinesses, digital platforms, agritech startups

VC ATTRACTIVENESS

Rarely standalone — high development value



Note: Unlike UC1–8, which each focus on a specific agricultural function, this use case addresses the interface and accessibility layer that conditions the real-world impact of all the others. It is included as a full use case because interface design and accessibility are areas of active startup innovation in Africa — and because failure to address them is one of the most consistent reasons technically sound solutions do not reach the farmers who need them.

Why it matters now

Many digital agricultural solutions exist across advisory, market access, and financial services — but adoption among smallholder farmers remains far below technical reach¹⁶. Heterogeneous technological maturity, strong language fragmentation across hundreds of local languages, and reliance on intermediaries create structural access barriers not resolved by improving model quality. Advances in generative AI and voice-based interfaces are opening new possibilities — but their impact in African agricultural contexts depends critically on local data and ecosystem integration, not on generic model capability.

KEY TAKEAWAYS

- The primary challenge is not generating agricultural insights but making them accessible across heterogeneous farming contexts — different devices, languages, literacy levels, and connectivity conditions.
- Generic AI models, including current LLMs, are insufficient for African agricultural contexts: local language data, locally relevant agronomic knowledge, and service ecosystem integration are required.
- The local language data infrastructure gap is as important as the agricultural data gap — and is a buildable asset, as demonstrated by Jokalante's Kallaama speech dataset initiative.

1. Problem addressed

A persistent gap exists between the availability of digital agricultural solutions and their actual use by farmers. This gap is reinforced by: variable technological maturity (from basic phones to smartphones); strong language fragmentation with limited localized interfaces; difficulty retaining users when value is not immediately tangible; and reliance on intermediaries to relay information. Even technically strong solutions remain underutilized if farmers cannot interact with them effectively — making accessibility a binding constraint on the impact of every other use case in this report.

2. AI / Data approach

This use case focuses on using AI to reduce friction between farmers and digital agricultural systems.

16. GSMA AgriTech (2020). "Digital Agriculture Maps." GSMA / IDH FarmFit. <https://www.gsma.com/digital-agriculture-maps/>. See also: GSMA AgriTech Accelerator Learnings Report (2025).

Key applications include: conversational interfaces (chatbots, voice assistants) delivering information in natural language; automated translation and localization into local languages; and simplification of complex outputs into actionable recommendations adapted to farmer literacy and device constraints. Voice-based systems are particularly relevant in African agricultural contexts: they enable information access through basic calls or audio messages without requiring smartphones or literacy, aligning with actual usage patterns in most rural farming communities. Recent LLM advances enable more natural conversational interactions – but in agricultural settings, generic models lack precision due to underrepresentation of local crops, practices, climate conditions, and languages. The key gap is not conversational capability but domain-specific, locally grounded knowledge – which must be built deliberately, not assumed to emerge from general-purpose models.

3. Startup approaches

Jokalante (Senegal) is a social enterprise founded in 2016 by Ndeye Amy Kebe, operating in Senegal and Gambia. Its multichannel platform – IVR, SMS, voice messages, and social media – delivers agricultural information in seven languages (Wolof, Serer, Pulaar, Diola, Mandinka, French, and English) and has reached 500,000 farmers over three years through B2B partnerships¹⁷ with national institutions including ANACIM and agricultural cooperatives. Since 2024, Jokalante has deployed Nafuore¹⁸, a generative AI voice chatbot adapting responses by weather conditions, soil type, and crop, trained on local agronomic knowledge bases and climate data. Jokalante won the Impact Award at Challenge+ Dakar 2025.

Beyond deploying AI tools, Jokalante has invested in building the language data infrastructure that models require. As lead of the Kallaama project (Lacuna Fund, 2023), it created 125 hours of transcribed agricultural speech across Wolof, Pulaar, and Serer¹⁹ – published open-source to support ASR and TTS development for languages with minimal digital resources. This makes a critical analytical point: the bottleneck for AI accessibility is not the conversational technology but the training data needed to make it locally functional. Funding the data infrastructure, not just the interface, is the relevant program and investment priority.

Interface approaches exist on a spectrum – SMS/USSD (high accessibility, limited complexity), mobile applications (richer features, higher device requirements), voice-based interfaces (higher usability, higher technical cost) – and are typically embedded in broader agricultural service platforms rather than deployed as standalone products.

4. Data challenges

Local agricultural practices and crops are underrepresented in AI training datasets globally. Many African languages have minimal digital resources for NLP development. Farmer-generated interaction data is irregular, unstructured, and difficult to standardize. Unlike other sectors where large structured datasets enable rapid model development, agricultural interface solutions require deliberate investment in local language corpora, agronomic knowledge bases, and contextual validation – investment that is rarely funded through commercial channels alone.

17. GSMA AgriTech (2023). "Jokalante, Senegal: Harnessing the Value of B2B Partnerships for Scaling." gsma.com/agritech/jokalante-senegal

18. Fund for Innovation in Development (FID). "Voice AI chatbot provides farming advice in Senegal." Project page (2024–2026). <https://fundinnovation.dev/en/projects/voice-ai-chatbot-provides-farming-advice-senegal>

19. Gauthier, E. et al. (2024). "Kallaama: A Transcribed Speech Dataset about Agriculture in the Three Most Widely Spoken Languages in Senegal." RAIL 2024, ACL Anthology: <https://aclanthology.org/2024.rail-1.2/>. Dataset: <https://www.openslr.org/151/> (Lacuna Fund). Full dataset: 125h – Wolof (55h), Pulaar (32h), Serer (38h).

5. Market and scaling dynamics

This use case is rarely viable as a standalone business. Its value lies in improving adoption and retention of agricultural services, enabling continuous farmer interaction, and facilitating direct data collection from users. These interface layers are typically funded and operated by platforms, agribusinesses, or development programs – with value captured indirectly by the services these tools make accessible.

Localization across multiple languages and farming systems generates significant marginal costs that constrain commercial scaling. Each new language or cultural context requires targeted adaptation investment – too small for commercial capital, too large for startups to absorb alone. This is precisely the structural gap that programs like AI4SU can address through open innovation and shared tooling.

6. Risks and limitations

- Limited performance of generic AI models due to lack of agricultural and local language training data.
- Risk of inaccurate or poorly contextualized recommendations – particularly damaging in low-literacy contexts where farmers cannot independently verify advice.
- Difficulty maintaining user engagement over time when value is not immediately tangible.
- Exclusion of farmers without access to basic connectivity.
- Risk of overestimating generative AI's near-term role where cost, connectivity, and local data availability are the primary limiting factors.

7. Data governance and compliance

This use case creates new and continuous data flows – voice interactions, usage patterns, conversational logs – raising questions about: collection and use of farmer-generated interaction data; dependency on external AI infrastructure and cloud providers; transparency of recommendations and embedded biases in training data; and data sovereignty when AI infrastructure is hosted outside deployment contexts. Voice interaction data may contain sensitive information beyond agricultural queries – raising user privacy considerations not always addressed in agricultural platform deployments.

8. Implementation considerations

Successful deployment requires strong localization (languages, agricultural practices, cultural context); adaptation to heterogeneous user profiles and device constraints; integration with existing agricultural services; and mechanisms to validate outputs and maintain farmer trust over time. The primary challenge is making AI systems genuinely usable, relevant, and trusted across diverse resource-constrained contexts – a design and partnership challenge as much as a technical one.

APPENDIX

A. African Agritech Startup Reference Landscape

This table provides a reference overview of startups and initiatives mentioned in the report. It is not intended as a comprehensive market map. Status and funding information are based on publicly available sources and should be treated as indicative.

Startup / Initiative	Country / Scope	Status	AI role	Sector focus	Cited in
Apollo Agriculture	Kenya, Zambia	Active	Embedded / Core	Agri-finance, inputs, advisory	UC1, UC6
FarmSawa*	Kenya	Active	Core	Inclusive credit scoring, agri-finance	UC1, UC2
Farmerline	Ghana, Côte d'Ivoire	Active	Embedded	Advisory, farmer data platform, market access	UC1, UC5
Pula	Kenya, pan-African	Active	Core	Agricultural insurance, risk modeling	UC2
ACRE Africa	Kenya, East & Central Africa	Active	Embedded	Index-based insurance, value-chain distribution	UC2
OKO	Mali, Côte d'Ivoire, Uganda	Active	Core	Digital crop insurance, climate risk	UC2
AgroCenta	Ghana	Active	Embedded	Market linkage, price transparency	UC3
agriBORA*	Kenya	Active	Embedded / Core	Market intelligence, warehousing, credit, market linkage	UC3, UC4
Twiga Foods	Kenya	Active	Embedded	Supply chain, logistics, aggregation, food distribution	UC3, UC4
Complete Farmer	Ghana	Active	Embedded	B2B sourcing, fulfillment, market access	UC4

Startup / Initiative	Country / Scope	Status	AI role	Sector focus	Cited in
Farm to Feed*	Kenya	Active	Embedded	Food waste reduction, surplus redistribution	UC4
AfriMash*	Nigeria, Ghana	Active	Embedded	Agri-e-commerce, input marketplace, customer intelligence	UC3, UC4
Ignitia	Ghana, Mali, Nigeria	Active	Core	Hyper-local weather intelligence, advisory	UC5
DigiCow*	Kenya	Active	Embedded	Dairy advisory, practice adoption	UC5
Rhea Soil*	Kenya	Active	Core	Soil intelligence, input optimization	UC5, UC6
Aerobotics	South Africa, pan-African	Active	Core	Crop analytics, aerial imagery, remote sensing	UC7
Tolbi*	Senegal / Côte d'Ivoire	Active	Core	Satellite monitoring, traceability, compliance	UC7, UC8
Jokalante	Senegal, Gambia	Active	Core	Voice, IVR, multilingual farmer advisory	UC9
Tulaa	Kenya	Inactive	Embedded	Input financing, market linkage, credit	Reference only
PlantVillage / Nuru	Kenya / global	Research initiative	Core	Mobile disease detection, computer vision	UC7 reference
Satelligence	Netherlands, global	Non-African HQ	Core	Geospatial monitoring, deforestation compliance	UC8 reference

B. Use cases mapping

The table below summarizes the nine use cases analyzed in Part 3, with indicative VC attractiveness ratings based on the analysis developed in each section.

AI4SU — AI FOR STARTUPS · DATA GOVERNANCE IN AFRICA

Where data & AI create value in African agriculture

From Use Cases to Real-World Impact — 9 use cases mapped across the agri value chain. Based on ecosystem analysis and 8 open innovation hackathons across Africa.

AI TYPE ● Predictive scoring ● Index & risk modeling ● Market intelligence ● Logistics optimization ● Remote sensing & CV ● Conversational AI AI4SU = program hackathon partner

#	USE CASE	AI TYPOLOGY	DATA & AI CHALLENGES	STARTUPS (ILLUSTRATIVE)	AI4SU HACKATHON	BUSINESS / VC REALITY
► FINANCE & RISK						
01	Access to finance Alternative credit scoring for smallholder farmers with no formal financial history. AGRITECH FINTECH	● Predictive scoring	- Thin-file data: no repayment history to train on at scale - Telecom & banking data inaccessible or fragmented - MVP data ≠ scalable data: each new market restarts accumulation - Consent & explainability in low-literacy contexts	KE Apollo Agriculture KE FarmSawa AI4SU GH Farmerline	FARMSAWA · KENYA · OCT 2025 Inclusive Credit Scoring for Smallholder Farmers	●●●●○ VC ATTRACTIVENESS Embedded models win. Scoring alone is not investable — it becomes a business when bundled with inputs, advisory, or lending infrastructure. DFI capital is active in this space.
02	Climate risk & insurance Index-based insurance products adapted to smallholder exposure to climate shocks. INSURTECH CLIMATECH	● Index & risk modeling	- Basis risk: index payout ≠ actual farm-level loss - Sparse ground-truth yield data for model calibration - Correlated climate shocks = solvency risk at scale - Trust fragile: one wrong payout kills renewal rates	KE Pula KE ACRE Africa ML/CI OKO KE FarmSawa AI4SU	FARMSAWA · KENYA · DEC 2025 Climate & Market Resilience	●●●●○ VC ATTRACTIVENESS Strategically important, commercially difficult. Viable only as a risk layer within broader platforms. Often requires blended finance or public risk-sharing to reach sustainability.





MARKETS & SUPPLY CHAINS

03 Market information & price transparency

Data-driven tools to reduce price asymmetry and improve farmer selling decisions.

MARKET INTEL

● Market intelligence

- Price is not a single value: negotiated, grade- & volume-dependent
- Traders withhold data — opacity is their competitive edge
- Farmer-side data: inconsistent units, irregular reporting
- Behavioral gap: info ≠ action (liquidity, perishability)

GH AgroCenta
KE agriBORA AI4SU
KE Twiga Foods
GH AfriMash AI4SU

AGRIBORA · KENYA · NOV-DEC 2025

Maize Price Intelligence for Smarter Selling

AFRIMASH · GHANA · OCT 2025

Customer Segmentation & Retention Intelligence

VC ATTRACTIVENESS

Information alone doesn't monetize. Value captured on execution (transactions, logistics, bundling). Systemic value: platforms progressively formalize informal market data.

04 Supply chain & market access

Data-enabled aggregation, logistics, and distribution to connect farmers to markets.

SUPPLY CHAIN FOODTECH

● Logistics optimization

- Unreliable supply data: volumes inconsistent, real-time hard
- Standardization prerequisite: units, quality grades, pricing
- AI is coordination & optimization — not prediction
- Each new market rebuilds data & relationships from scratch

KE Twiga Foods
KE agriBORA AI4SU
GH Complete Farmer
KE Farm to Feed AI4SU

FARM TO FEED · KENYA · NOV-DEC 2025

Produce Recommendation Systems for Food Waste Reduction

VC ATTRACTIVENESS

Execution over algorithms. Thin margins, capital intensity. Defensibility from operational density & network control, not data models. Replication across markets is slow.

PRODUCTIVITY & SUSTAINABILITY

05 Agronomic advisory

Multi-variable decision support for planting, crop management, and risk anticipation.

ADVISORY CLIMATECH

● Predictive scoring

● Remote sensing & CV

- No ground-truth feedback loops from farmers
- Satellite data: resolution vs. cost trade-off
- Adoption gap: recommendations not acted on
- Attribution: hard to isolate advisory's impact on yields

GH Farmerline
GH/ML Ignitia
KE DigiCow AI4SU
KE Rhea Soil AI4SU

DIGICOW · KENYA · JAN-FEB 2026

Predicting Practice Adoption in Smallholder Farming

VC ATTRACTIVENESS

Easy to build, hard to monetize. Farmers rarely pay for advice. Value captured by input suppliers & lenders. Works when embedded in a platform with an existing commercial relationship.

06 Input optimization & sustainable farming

Data-driven input recommendations linked to supply chains and compliance requirements.

AGRONOMY SUSTAINABILITY

● Predictive scoring

- Soil data costly & scarce — sampling economics don't scale
- Plot-level variability limits model generalization
- Conflict of interest: advisory linked to input sales
- Compliance costs unevenly distributed across actors

KE Rhea Soil AI4SU
KE Apollo Agriculture
KE Farm to Feed

RHEA SOIL · KENYA · FEB-MAR 2026

Soil Nutrient Prediction for Sustainable Farming

VC ATTRACTIVENESS

Clearer economic logic than advisory. Value flows to input suppliers & agribusinesses more than farmers directly. Sustainability compliance is becoming a commercial driver, not just an ESG signal.





07 Crop monitoring & disease detection

Computer vision and remote sensing to detect crop risks early at field or landscape scale.

REMOTE SENSINGPRECISION AG

Remote sensing & CV

- Labeled dataset gap: domain expertise + continuous update needed
- Open satellite data: resolution constraints for plot-level use
- Detection ≠ impact: response pathway must exist to create value
- Generic LLMs insufficient: local crops underrepresented in training

SA Aerobotics
CI Tolbi AI4SU

TOLBI · CÔTE D'IVOIRE · APR-JUN 2025
Pixel-Level Crop Classification – 866 participants

VC ATTRACTIVENESS

Technically central but standalone is weak. AI genuinely useful here (CV, remote sensing) – but detection without accessible response has no farm-level impact. Best embedded in compliance or advisory models.

COMPLIANCE & MARKET ACCESS

08 Traceability & regulated market access

AI-enabled supply chain traceability for EUDR compliance and international market access.

TRACEABILITYCOMPLIANCE

Remote sensing & CV

Logistics optimization

- Data fragmented across actors with no standardization
- Open satellite: resolution gap for plot-level verification
- Fraud risk in declarative systems without independent audit
- Data sovereignty: compliance data flows to importers, not producers

CI Tolbi AI4SU

No dedicated hackathon – UC7 Tolbi crop classification models directly support compliance verification.

VC ATTRACTIVENESS

Regulation-driven demand is reliable. B2B, higher WTP than farmer-facing models. Strategic risk: African startups must build sovereign infrastructure, not just compliance data pipelines for European buyers. (US\$11B export risk figure is a modeled upper bound.)

CROSS-CUTTING ENABLER

09 AI accessibility & farmer interfaces

Voice, IVR, and conversational AI to make agricultural services accessible across all literacy levels and devices.

DIGITAL INCLUSION

Conversational AI

- Local language data gap: training corpora barely exist
- Generic LLMs fail: local crops & practices absent from training
- Localization cost per language/context is high & rarely funded commercially
- Connectivity exclusion: not all farmers reached even by voice

SN Jokalante AI4SU

500K farmers · 7 languages
Nafore AI chatbot · Kallaama: 125h speech data (Wolof, Pulaar, Sereer)

No dedicated hackathon – cross-cutting enabler for all other use cases.

VC ATTRACTIVENESS

Rarely standalone – high development value. Impact multiplier for all other UCs. Funded through platforms & development programs. Language data infrastructure is the real gap to fill.

Part of Data Governance in Africa, supported by @AfricanUnion, @EU_Commission, @BMZ_Bund and implemented by @HAUSfi, @ESTDEV, @GIZAFfricanUnion, @Enabel_Belgium, @expertisefrance and #DigitalAfrica

AI4SU (AI for Startups) is a component of the Data Governance in Africa EU initiative, implemented by Digital Africa in collaboration with Expertise France. · Startups listed are illustrative examples based on publicly available information as of early 2026 and activities run during the program. · VC attractiveness scores are qualitative assessments based on ecosystem analysis; they do not constitute investment advice. · Geographical scope: Senegal · Côte d'Ivoire · Ghana · Tanzania · Rwanda · Kenya · South Africa · Botswana.

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